Advanced Macroeconomics

I. Foundations of Dynamic Macroeconomic Modeling
   1. Introduction
   2. History of Macroeconomics
   3. Static General Equilibrium Models
   4. Saving and Investment in a Two-Period Model

II. Long-run Economic Growth

III. Short-run Fluctuations

IV. Applications

Can Tomorrow's Good News Change Consumption Today?

A Future Productivity Improvement

Suppose households learn today that firms will become 10% more productive in period 2

$$A_1 = 1, \qquad A_2 = 1.1$$

Productivity in period 1 remains unchanged; the improvement is known with certainty

- What happens to consumption today (c_1) and consumption tomorrow (c_2)?
- Will households save and invest more or less in period 1?
- Can you explain your predictions using economic intuition?

4. Saving and Investment in a Two-Period Model

Two-Period Model
Sensitivity Analysis
Reaction to Shocks
Consumption and Labor Supply in a Two-Period Competitive Model
Summary and Literature

Two-Period Set-up

- Fisher (1930)
- Period 1: the present
- Period 2: the future
- Perfect foresight
- Households consume, save and invest: in period 2, the capital stock is equal to the initial endowment plus investment from the first period

Time Preference

- Time preference: Households value utility received today more highly than the same amount of utility received in the future
- Discounting: One unit of utility in period 2 receives a weight of β, compared with a weight of one in period 1
- β is called the discount factor ($0 < \beta < 1$)
- The discount factor depends on the time preference rate ($\rho > 0$):

$$\boxed{\beta = \frac{1}{1+\rho}}$$

Utility Function and Budget Constraint in the Two-Period Model

- Intertemporal utility function

$$U(c_1, c_2) = u(c_1) + \frac{1}{1+\rho} u(c_2)$$

- Budget constraint period 1

$$k_1 + r_1 k_1 + w_1 n_1 - c_1 = k_2$$

- Budget constraint period 2

$$c_2 = (1 + r_2) k_2 + w_2 n_2$$

- Assume $k_1 = 1$ and $n_1 = n_2 = 1$ and combine the two constraints

$$c_1 + \frac{1}{1+r_2} c_2 = (1 + r_1) + w_1 + \frac{1}{1+r_2} w_2$$

Utility Maximization in the Two-Period Model

- Lagrange function

$$\mathcal{L} = u(c_1) + \frac{1}{1+\rho}u(c_2) - \lambda\left(c_1 + \frac{1}{1+r_2}c_2 - (1+r_1) - w_1 - \frac{1}{1+r_2}w_2\right)$$

- First-order conditions

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial c_1} &= u'(c_1) - \lambda = 0 \quad \Rightarrow \quad u'(c_1) = \lambda \\
\frac{\partial \mathcal{L}}{\partial c_2} &= \frac{1}{1+\rho}u'(c_2) - \frac{\lambda}{1+r_2} = 0 \quad \Rightarrow \quad \frac{1}{1+\rho}u'(c_2) = \frac{\lambda}{1+r_2}
\end{aligned}$$

- Euler equation for consumption

$$\frac{1}{1+\rho}\frac{u'(c_2)}{u'(c_1)} = \frac{1}{1+r_2} \quad \Leftrightarrow \quad \frac{u'(c_2)}{u'(c_1)} = \frac{1+\rho}{1+r_2}$$

Euler Equation for Power Utility Function

- Power utility (constant elasticity of intertemporal substitution of consumption $1/\theta$)

$$u(c) = \begin{cases} \dfrac{c^{1-\theta} - 1}{1-\theta}, & \theta \neq 1, \\ \ln c, & \theta = 1. \end{cases}$$

where

$$u'(c) = c^{-\theta}, \qquad u''(c) = -\theta c^{-\theta-1}, \qquad \frac{-cu''(c)}{u'(c)} = c\frac{\theta c^{-\theta-1}}{c^{-\theta}} = \theta$$

- EULER equation

$$\beta \frac{u'(c_2)}{u'(c_1)} = \beta \left(\frac{c_1}{c_2}\right)^{\theta} = \frac{1}{1+r_2}$$

Profit Maximization

- Firms maximize the present value of profits

$$\mathcal{P} = A_1 F(k_1, n_1) - r_1 k_1 - w_1 n_1 + \frac{1}{1 + r_2} \left(A_2 F(k_2, n_2) - r_2 k_2 - w_2 n_2 \right)$$

- First-order conditions

$$\begin{aligned} r_1 &= A_1 F_k(k_1, n_1) \\ r_2 &= A_2 F_k(k_2, n_2) \\ w_1 &= A_1 F_n(k_1, n_1) \\ w_2 &= A_2 F_n(k_2, n_2) \end{aligned}$$

General Equilibrium in the Two-Period Model

- Initial capital stock (k_1) and total factor productivity (A_1, A_2) are exogenous
- 9 endogenous variables: y_1, y_2, c_1, c_2, r_1, r_2, w_1, w_2, k_2
- 9 equations
  - 2 production functions (for period 1 and for period 2, respectively)
  - 2 budget constraints (for period 1 and for period 2, respectively)
  - Euler equation for consumption
  - 4 profit maximizing first-order conditions
- Solve numerically for general equilibrium: `twoperiodmodel.m`

Hands-on Exercise 1: Time Preference and Saving

Your Task

Open `twoperiodmodel.m` in Matlab or Octave and investigate how greater impatience affects consumption and capital accumulation

1. **Compute the baseline equilibrium.** Run the script with $\rho = 0.05$. Record c_1, c_2, k_2, and r_2.
2. **Increase impatience.** Change the time preference rate

   ```
   params.rho = 0.10;
   ```

   Run the script again
3. **Compare the equilibria.** How do consumption in both periods, the capital stock k_2, and the interest rate r_2 change?
4. **Explain the results.** Why does a change in time preference affect saving and investment? Use the Euler equation

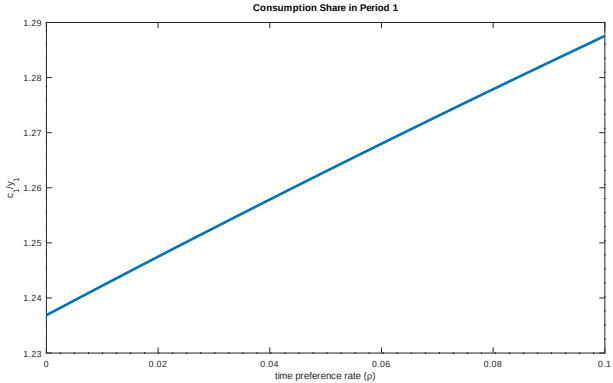
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Sensitivity to Time Preference

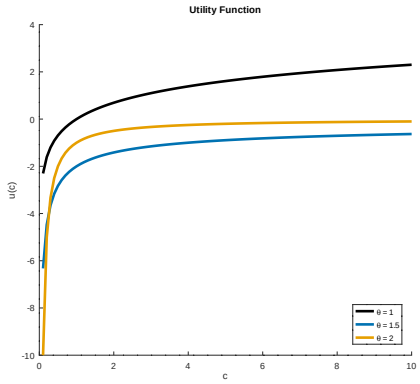
(`twoperiodsensitivity.m`)

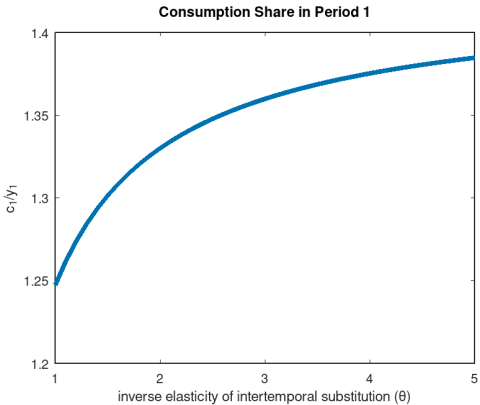
If the time preference rate ρ increases, what happens to the consumption share c_1/y_1?



The Higher the Inverse Elasticity of Intertemporal Substitution, the Higher Period 1 Consumption

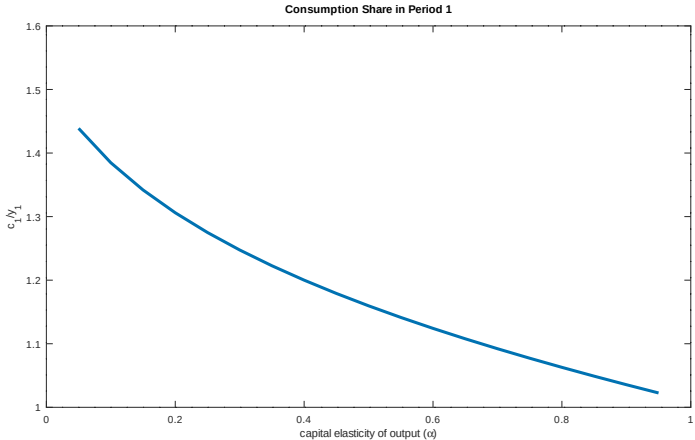
(`twoperiodsensitivity.m`)





The Higher Capital Elasticity of Output, the Lower Period 1 Consumption

(`twoperiodsensitivity.m`)

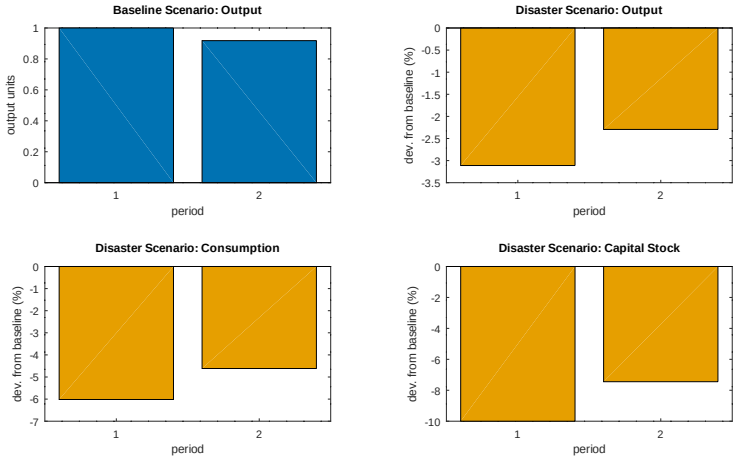


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Effects of a Natural Disaster: Rebuilding the Capital Stock

(`twoperioddisaster.m`, baseline: $k_1 = 1$, disaster: $k_1 = 0.9$)



Hands-on Exercise 2: Productivity Shocks

Your Task

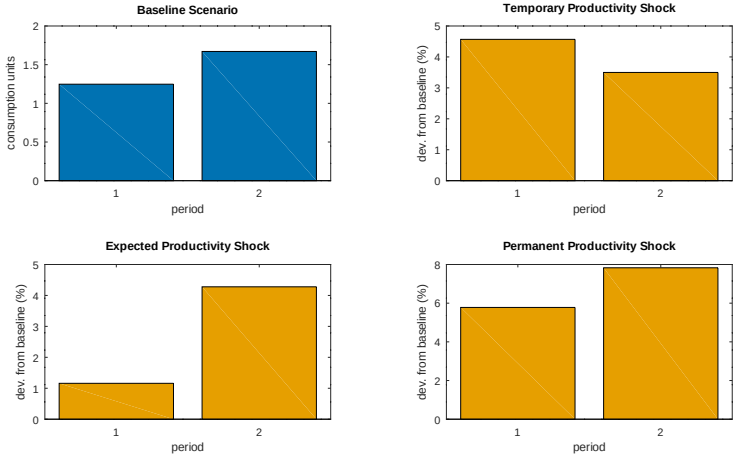
Open `twoperiodproductivity.m` in Matlab or Octave and compare how different productivity shocks affect consumption and saving

1. Predict the effects. Consider three scenarios with a 10% productivity increase
   - Temporary: $A_1 = 1.1$, $A_2 = 1$
   - Anticipated: $A_1 = 1$, $A_2 = 1.1$
   - Permanent: $A_1 = A_2 = 1.1$

   How should c_1 and c_2 respond?
2. Run the simulations. Execute `twoperiodproductivity.m` and examine the results
3. Compare the equilibria. Which shock has the largest effect on c_1? How do the responses of c_2 differ?
4. Explain the results. Why can an anticipated productivity increase affect current consumption? What role does consumption smoothing play?

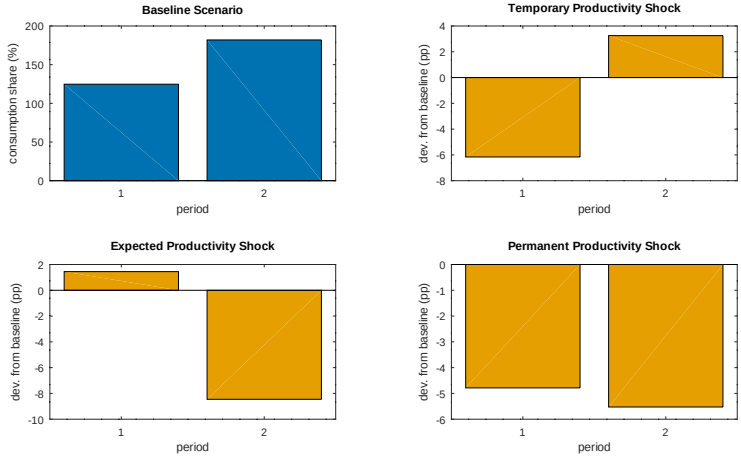
Productivity Shocks and Consumption Smoothing I

(`twoperiodproductivity.m`, Shock Size 10%)



Productivity Shocks and Consumption Smoothing II

(`twoperiodproductivity.m`, Shock Size 10%)



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Does Higher Productivity Increase Employment?

Recall from Chapter 3

With logarithmic consumption utility, a productivity improvement raised output, consumption, and the real wage, but employment remained unchanged

Now Consider a Two-Period Economy

Households choose consumption and labor supply in both periods and can borrow or lend
Suppose productivity increases only in period 2

$$A_1 = 1, \qquad A_2 = 1.1$$

1. What do you expect to happen to employment in periods 1 and 2?
2. Which income and substitution effects might influence labor supply?
3. Does the result from Chapter 3 necessarily carry over to this economy?

Utility Function and Budget Constraint

- Intertemporal utility function

$$U = \frac{c_1^{1-\theta}}{1-\theta} - \frac{n_1^{1+\gamma}}{1+\gamma} + \frac{1}{1+\rho}\left(\frac{c_2^{1-\theta}}{1-\theta} - \frac{n_2^{1+\gamma}}{1+\gamma}\right)$$

- Households can borrow and lend on a competitive financial market at interest rate r
- Budget constraint period 1

$$w_1 n_1 - c_1 = a_2$$

- Budget constraint period 2

$$c_2 = (1+r)a_2 + w_2 n_2$$

- Combine the two constraints: intertemporal budget constraint

$$c_1 + \frac{c_2}{1+r} = w_1 n_1 + \frac{w_2 n_2}{1+r}$$

Utility Maximization

- Lagrangian

$$\mathcal{L} = \frac{c_1^{1-\theta}}{1-\theta} - \frac{n_1^{1+\gamma}}{1+\gamma} + \frac{1}{1+\rho}\left(\frac{c_2^{1-\theta}}{1-\theta} - \frac{n_2^{1+\gamma}}{1+\gamma}\right) - \lambda\left(c_1 + \frac{c_2}{1+r} - w_1 n_1 - \frac{w_2 n_2}{1+r}\right)$$

- First-order conditions

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial c_1} &= c_1^{-\theta} - \lambda = 0 \quad \Rightarrow \quad c_1^{-\theta} = \lambda \\
\frac{\partial \mathcal{L}}{\partial c_2} &= \frac{c_2^{-\theta}}{1+\rho} - \frac{\lambda}{1+r} = 0 \quad \Rightarrow \quad \frac{c_2^{-\theta}}{1+\rho} = \frac{\lambda}{1+r} \\
\frac{\partial \mathcal{L}}{\partial n_1} &= -n_1^{\gamma} + \lambda w_1 = 0 \quad \Rightarrow \quad n_1^{\gamma} = \lambda w_1 \\
\frac{\partial \mathcal{L}}{\partial n_2} &= -\frac{n_2^{\gamma}}{1+\rho} + \frac{\lambda}{1+r} w_2 = 0 \quad \Rightarrow \quad \frac{n_2^{\gamma}}{1+\rho} = \frac{\lambda}{1+r} w_2
\end{aligned}$$

Household Equilibrium Conditions

- Consumption EULER equation

$$\frac{1}{1+\rho}\left(\frac{c_2}{c_1}\right)^{-\theta} = \frac{1}{1+r}$$

- Labor supply EULER equation

$$\frac{1}{1+\rho}\left(\frac{n_2}{n_1}\right)^{\gamma} = \frac{1}{1+r}\frac{w_2}{w_1}$$

- Labor supply in period 1

$$\frac{n_1^{\gamma}}{c_1^{-\theta}} = w_1$$

- Labor supply in period 2

$$\frac{n_2^{\gamma}}{c_2^{-\theta}} = w_2$$

Profit Maximization

- Firms maximize the present value of profits

$$\mathcal{P} = F(n_1) - w_1 n_1 + \frac{1}{1+r}\left(F(n_2) - w_2 n_2\right)$$

- First-order conditions

$$\begin{aligned} w_1 &= F_n(n_1) \\ w_2 &= F_n(n_2) \end{aligned}$$

- For linear production $y_t = A_t n_t$:

$$\begin{aligned} w_1 &= A_1 \\ w_2 &= A_2 \end{aligned}$$

General Equilibrium in the Two-Period Model

- Productivity (A_1, A_2) is exogenous
- 9 endogenous variables: y_1, y_2, c_1, c_2, r, w_1, w_2, n_1, n_2
- 9 equations
  - 2 resource constraints: $y_1 = c_1$ and $y_2 = c_2$
  - Euler equation for consumption
  - 2 labor supply equations (for period 1 and period 2, respectively)
  - 2 profit maximizing first-order conditions (labor demand)
  - 2 production functions (for period 1 and for period 2, respectively)
- Solve numerically for general equilibrium: `twoperiodlabourmodel.m`

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Summary: Saving and Investment in a Two-Period Model

- Intertemporal optimization: Households allocate consumption across periods; the Euler equation links consumption decisions to time preference and the real interest rate
- Saving and investment: Lower consumption today allows more capital accumulation and production tomorrow
- Productivity shocks: Their effects depend on timing and persistence; households adjust saving to smooth consumption
- Endogenous labor supply: Employment responds to income and substitution effects; the outcome depends on preferences and technology

Back to the Opening Puzzle

Can tomorrow's good news change consumption today?

Yes! In the saving-and-investment model, higher future productivity raises consumption today, even if current productivity is unchanged; households reduce saving and investment.

Outlook

- The two-period model of savings and investment is the basis of most one-sector models of economic growth (Chapters 5 to 7) and of competitive models of aggregate fluctuations (Chapter 8)
- The two-period model with endogenous labor supply is the basis of DSGE models of aggregate fluctuations (Chapters 9 and 10)

Literature

 Alogoskoufis, George (2019): Dynamic Macroeconomics, MIT Press, Chapter 2

 Fisher, Irving (1930): The Theory of Interest, Macmillan

 Friedman, Milton (1957): A Theory of the Consumption Function, Princeton University Press